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Preparation and application of a novel biomass-based amphoteric retanning agent with the function of reducing free formaldehyde in leather

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ABSTRACT

Non-toxic and biodegradable retanning agents have received increasing attention over recent years driven by the tough environmental targets for leather cleaner manufacturing. Herein, a novel biomass-based amphoteric retanning agent, which can reduce free formaldehyde significantly, was prepared using hydrolyzed collagen and oxidized valonia extract as the raw materials. In this investigation, collagen was derived from chromium-containing waste leather by *in situ* hydrolysis, and oxidized valonia extract was obtained after laccase catalysis of valonia extract. A Schiff's base reaction occurred between the laccase-catalyzed valonia extract and hydrolyzed collagen. Notably, the leather retanned using the oxidized valonia extract modified collagen retanning agent was comparable and surpassed a market glutaraldehyde modified collagen retanning agent exhibiting superior shrinkage temperature, thickness, mechanical properties, dyeing properties and free formaldehyde removal function (free formaldehyde removal rate is 70.2%). This study provides a method for preparing an outstanding multifunctional non-toxic amphoteric retanning agent and offers a method for sustainable development of leather.

1. Introduction

Retanning process can change the properties of leather significantly and plays an important role in leather manufacturing (Wang et al., 2019a,b). The leather industry uses many common inorganic and organic retanning agents. Inorganic salt retanning agents include basic chromium, zirconium, and aluminum salts, while organic retanning agents include vegetable tanning agent, synthetic tanning agent, aldehyde tanning agent, oil tanning agent, and resin tanning agent (Sun et al., 2018; Tasca and Puccini, 2019). However, there is a scarcity of reports on the preparation of multifunctional retanning agents with improved leather shrinkage temperature, thickening rate, mechanical properties, assisting dyeing performance, and more importantly, the ability to remove free formaldehyde in the leather manufacturing process.

China is the largest producer of leather worldwide and could generate approximately 1.4 million tons of solid waste per year, of which 280,000 tons of solid waste contain chromium resource (Jiang et al., 2016). Among the chromium-containing solid wastes, collagen accounts for approximately 90%. If this waste is not treated properly, precious natural collagen resources will be wasted, causing harm to the environment and society (Hashem and Nur-A-Tomal, 2018). With the increasing global attention towards environmental issues, more attention should be directed to utilize solid leather waste as a valuable resource (Konikkara et al., 2016; Yang et al., 2019). Surprisingly, due to its biomimetic structure to leather collagen, the collagen extracted from solid leather waste can easily fuse into the collagen fiber system in the skin, further

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